



EE214 Digital Circuits Laboratory

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Problem set: 1

Date: August 7, 2026

Designing Circuits using Universal Gate NOR

Instructions:

1. NOR is a universal gate.
2. For writing VHDL description, use the NOR gate provided in Gates.vhdl (which has been provided in [Resource Files](#)).
3. Do pen paper design, use proper labeling for each wire and use the same labels in the VHDL code.
4. Perform RTL simulation using the provided testbench and tracefile.

Problem Statement:

1. Design and describe AND gate, XOR gate, OR Gate using NOR gate in VHDL using Structural modeling.
2. Design Half Subtractor and Full Subtractor using NOR gates (pen-paper design).
3. Describe Half Subtractor(X-Y) and Full Subtractor(X-Y-B) using NOR gates in VHDL using Structural modeling (use the XOR gate that you have designed using NOR gates). Here X, Y are input bits and borrow bit is represented by input bit B for Full Subtractor.
4. Verify the working of your design by performing RTL simulation using the given tracefile and testbench.

NOTE:

- TRACEFILE for AND gate: [AND_Tracefile](#)
Format: InputX1 X0 OutputY0 MASK1
- TRACEFILE for OR gate: [OR_Tracefile](#)
Format: InputX1 X0 OutputY0 MASK1
- TRACEFILE for XOR gate: [XOR_Tracefile](#)
Format: InputX1 X0 OutputY0 MASK1
- TRACEFILE for Half Subtractor: [Half_Subtractor_Tracefile](#)
Format: Input{X Y} Output{Difference Borrow} MASK{1 1}
- TRACEFILE for Full Subtractor : [Full_Subtractor_Tracefile](#)
Format: Input{X Y B} Output{Difference Borrow} MASK{1 1}